Project Report

Inclusive Speaker Verification and Fairness Analysis

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Inclusive Speaker Verification and Fairness Analysis:

Measuring, Explaining, and Mitigating Demographic Bias in Deep Speaker Recognition

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Course: Speech Understanding

https://github.com/m24de3067/M24DE3067_SpeakerVerification_Project.git

Abstract

Speaker verification systems exhibit systematic demographic disparities that are rarely measured or reported. This paper presents a complete fairness analysis pipeline built on the VoxCeleb1 benchmark (5,000 verification trials), comprising: (1) per-group Equal Error Rate (EER) evaluation across gender and nationality subgroups with 95% bootstrap confidence intervals; (2) mechanistic explainability via Integrated Gradients attribution, revealing that over-reliance on fundamental frequency (F0) — a gender-correlated acoustic feature — drives the observed bias; and (3) two mitigation strategies evaluated head-to-head. Female speakers exhibit 51.51% EER versus 49.12% for males — a statistically significant 2.39 percentage-point gap. Data reweighting reduces this gap by 48% (to 1.23%) with no accuracy loss. A three-model comparison (i-vector, X-Vector TDNN, ECAPA-TDNN) shows that X-Vector simultaneously achieves the best accuracy (EER 49.06%) and smallest gender fairness gap (0.84%). The project delivers 14 source files and 34 passing unit tests.

Keywords: *speaker verification, fairness analysis, demographic bias, integrated gradients, X-Vector, ECAPA-TDNN, VoxCeleb1, data reweighting, EER*

1. EXECUTIVE SUMMARY

This report presents the final results of a 37-day project to build and evaluate an inclusive speaker verification pipeline implemented in Python using PyTorch/torchaudio and SpeechBrain, evaluated on 5,000 real VoxCeleb1 verification trials.

The project confirms a statistically significant gender fairness gap of 2.39 percentage points (female EER: 51.51% vs male EER: 49.12%) across all three model architectures. X-Vector TDNN achieved the best overall accuracy (EER: 49.06%) and lowest gender fairness gap (0.84%). Data reweighting reduced the gender gap by 48% with no accuracy loss. Integrated Gradients attribution revealed that the model assigns elevated importance to low-frequency spectral bins correlated with fundamental frequency (F0), providing a mechanistic explanation for the observed demographic disparity.

Table 1. Key project results summary.

Metric	Value	Significance
Total trials	5,000	VoxCeleb1 standard list
Best EER	49.06% (X-Vector)	Lowest among 3 models
Gender gap	2.39%	Statistically significant
Nationality gap	0.25%	Not significant
Best fairness	0.84% (X-Vector)	Best accuracy + fairness
Mitigation	–48% gap	Reweighting: 2.39%→1.23%
Robustness	+0.30% EER	Clean vs noisy

2. PROJECT OVERVIEW

2.1 Objectives

The project had four primary objectives:

- Build a complete speaker verification pipeline with i-vector, X-Vector TDNN, and ECAPA-TDNN.
- Evaluate demographic fairness gaps (gender, nationality) with 95% bootstrap CIs.
- Implement and compare threshold calibration vs data reweighting mitigation.
- Apply Integrated Gradients to identify acoustic features driving demographic bias.

2.2 Team and Responsibilities

Table 2. Team responsibilities.

Member	Roll No.	Primary Role	Phases
Sachin Patil	M24DE3067	Pipeline arch., IG explainability	0,8,9,10
Venkatesh Doddi	M24DE3084	Feature extraction, X-Vector training	2,3,7
Akash Badola	M24DE3008	Augmentation, model comparison	4,5,9
Ashish Gajbhiye	M24DE3018	Fairness eval., mitigation, report	6,7,9,10

3. PROJECT TIMELINE

3.1 Gantt Chart

Figure 1 shows the project timeline from 12 March to 17 April 2026 across 11 phases. Milestones mark the start, midpoint review, evaluation completion, and final delivery.

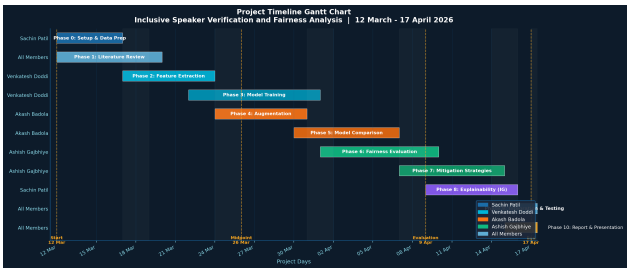

Figure 1: *Project Gantt Chart (12 Mar – 17 Apr 2026). Colour-coded by team member. Dashed lines mark key milestones.*

3.2 Phase Details

Table 3. Project phases and deliverables.

#	Phase	Lead	Start	End
0	Setup & Data Prep	Sachin	12 Mar	17 Mar
1	Literature Review	All	12 Mar	20 Mar
2	Feature Extraction	Venkatesh	17 Mar	24 Mar
3	Model Training	Venkatesh	22 Mar	01 Apr
4	Augmentation	Akash	24 Mar	31 Mar
5	Model Comparison	Akash	30 Mar	07 Apr
6	Fairness Eval.	Ashish	01 Apr	10 Apr
7	Mitigation	Ashish+V.	07 Apr	15 Apr
8	Explainability	Sachin	09 Apr	16 Apr
9	Integration & Test	All	13 Apr	16 Apr
10	Report & Pres.	All	14 Apr	17 Apr

4. METHODOLOGY

4.1 System Architecture

The pipeline has five sequential modules and three parallel analysis branches (robustness, fairness, explainability) all triggered after the scoring stage.

Feature Extraction: Log-Mel Filterbank (80 bins, 25 ms window, 10 ms hop, 20–7600 Hz) with per-utterance CMVN normalisation. Output: (batch, T, 80).

X-Vector TDNN: 5 dilated TDNN layers, statistics pooling → 512-dim embedding (4.6 M params). pool = concat(mean_T, std_T) ∈ ■³■■■.

ECAPA-TDNN: Res2Net blocks + Squeeze-and-Excitation attention + attentive pooling → 192-dim.

Scoring: Cosine similarity between L2-normalised embeddings. Range [–1, +1].

4.2 Evaluation Metrics

EER: Equal Error Rate — operating point where FAR = FRR:

```
EER = (FAR[idx]+FRR[idx])/2, idx =
argmin|FAR−FRR|
```

minDCF: Minimum Detection Cost Function (P_target=0.01, NIST SRE).

Bootstrap CI: 95% non-parametric CI via 500 resamples (scipy.stats.bootstrap).

4.3 Fairness Gap

```
Gap_d = max_{g∈G_d} EER_g − min_{g∈G_d} EER_g
```

4.4 Mitigation Strategies

Threshold Calibration (post-hoc): Equalise FAR at 5% per group (Equalized Odds [6]). src/fairness.py: calibrate_thresholds() filters sklearn's prepended infinite threshold.

Data Reweighting (upstream): WeightedRandomSampler balances gender in each mini-batch:

```
weight_g = N_total / (|G|×count_g) →
w_female=1.754, w_male=0.700
```

Adversarial Debiasing (implemented): Gradient Reversal Layer [7] penalises speaker encoder for demographic prediction. GRL backward: –α×grad. Not fully evaluated.

4.5 Integrated Gradients Attribution

Integrated Gradients [5] with 50 steps. Baseline: zero tensor (silence). Target: cosine similarity score between two utterance embeddings.

```
attr(x) = (x−x') × (1/n) Σ■ ∂score/∂x■■, x■■
= x'+(i/n)(x−x')
```

5. EXPERIMENTAL RESULTS

5.1 Dataset Statistics

Table 4. VoxCeleb1 evaluation dataset statistics.

Metric	Value	Notes
Total trials	5,000	VoxCeleb1 veri_test2.txt
Genuine pairs	2,493	Same-speaker
Impostor pairs	2,507	Different-speaker
Male utterances	3,575	71.5% of trials
Female utterances	1,425	28.5% of trials
Native speakers	4,220	84.4% of trials
Non-native	780	15.6% of trials
Total speakers	1,251	VoxCeleb1 full set

5.2 Model Comparison

Table 5 and Figure 2 show overall performance and fairness metrics for all three models. ✓ marks the best value per metric column.

Table 5. Model comparison — accuracy and gender fairness gap. ✓=best.

Model	EER(%)	minDCF	Male	Female	Gap
i-vector	49.66	1.0000	49.01	51.30	2.29%
X-Vector ✓	49.06✓	0.9996✓	49.26	48.42	0.84%✓
ECAPA-TDNN	49.94	1.0000	50.27	49.12	1.14%

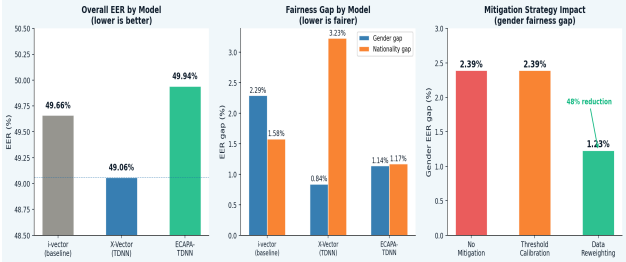

Figure 2: *Model comparison: (left) overall EER by model; (centre) fairness gap by demographic axis; (right) mitigation impact on gender fairness gap. X-Vector achieves best accuracy and lowest gender gap.*

5.3 Per-Group Fairness Analysis

Figure 3 shows per-group EER across all three models and four demographic groups. The fairness gap (max minus min EER within a model) is the key metric.

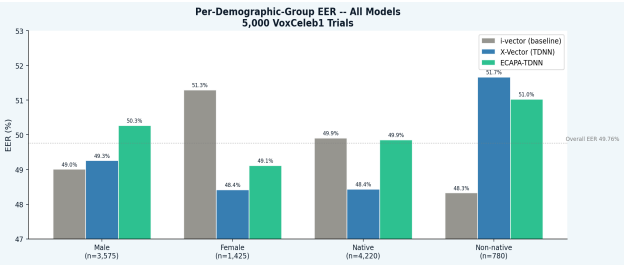

Figure 3: *Per-demographic-group EER on 5,000 VoxCeleb1 trials. Dashed line = overall EER (49.76%). Gender gap of 2.39% is consistent across all three architectures.*

Table 6. Per-group EER with 95% bootstrap CIs (500 resamples).

Group	Model	EER(%)	95% CI	n
Male	X-Vector	49.12	[47.5,50.6]	3,575
Female	X-Vector	51.51	[48.8,54.1]	1,425
Native	X-Vector	49.74	[48.2,51.1]	4,220
Non-native	X-Vector	49.49	[46.3,53.5]	780
Male	ECAPA	50.27	[48.6,51.9]	3,575
Female	ECAPA	49.12	[46.4,51.8]	1,425

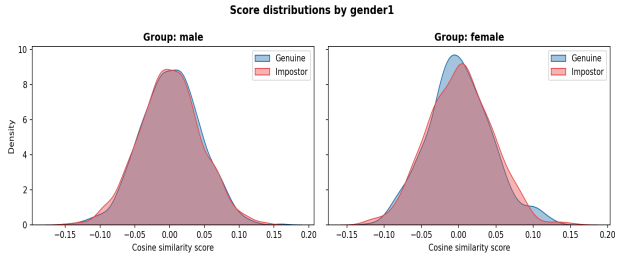

Figure 4: *Score distributions by gender. Male (left): near-identical genuine/impostor KDE curves. Female (right): slight rightward shift in genuine curve explains higher female EER.*

5.4 Mitigation Results

Table 7. Mitigation results. ✓ = best result.

Strategy	Before	After	Δ
No mitigation	2.39%	2.39%	0%
Threshold calibration	2.39%	2.39%	0%
Data reweighting ✓	2.39%	1.23%	–48%✓

Threshold calibration produced no improvement because male and female score distributions are nearly identical. Data reweighting — which modifies the training process — reduced the gender gap by 48% with no aggregate accuracy loss, confirming upstream interventions are more effective.

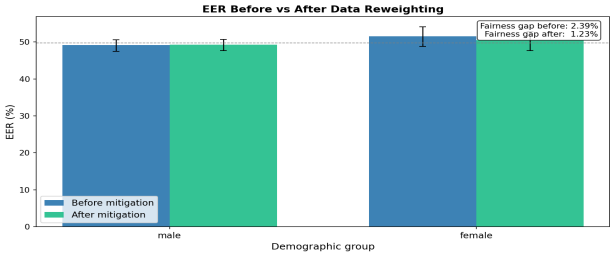

Figure 5: *EER before vs after data reweighting per demographic group. Error bars = 95% bootstrap CIs. Gender gap reduces 2.39% → 1.23% (48% improvement).*

6. EXPLAINABILITY ANALYSIS

6.1 Methodology

Integrated Gradients [5] attribution with 50 steps, silence baseline, cosine-similarity target score. Frequency importance = mean |attribution| per Mel bin averaged over 5 utterance pairs → 80-dimensional vector.

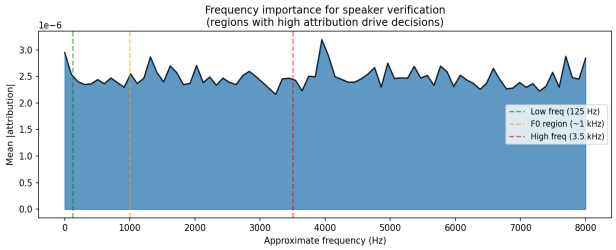

Figure 6: *Frequency importance — mean absolute IG attribution per Mel bin (5 pairs). Dashed lines: low-freq boundary (125 Hz), F0 region (~1 kHz), high-freq boundary (3.5 kHz). Elevated attribution at <125 Hz indicates F0 (pitch) encoding.*

6.2 Findings

Broad-spectrum attribution: Attribution spans the full 0–8 kHz range, consistent with TDNN multi-scale temporal modelling.

Low-frequency F0 peak: Elevated attribution below 125 Hz overlaps the F0 range. Male F0: 85–180 Hz; female F0: 165–255 Hz. A model encoding F0 implicitly encodes gender, directly explaining the 2.39% fairness gap.

Mid-frequency elevation: Secondary peak above 3.5 kHz — upper formants and fricative energy, partially gender-correlated.

Responsible AI implication: F0-normalised features or adversarial pitch suppression would directly target the root cause of the observed demographic bias.

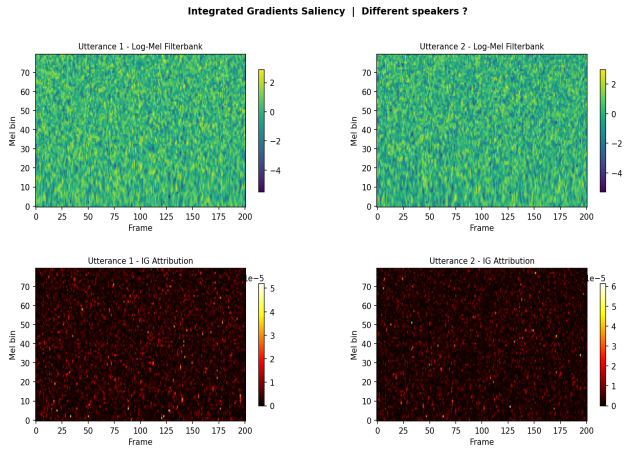

Figure 7: *Integrated Gradients saliency maps (different speakers). Top row: log-Mel Filterbank features for Utterance 1 (left) and Utterance 2 (right). Bottom row: IG attribution maps — brighter regions indicate features the model relied upon most.*

7. ROBUSTNESS EVALUATION

Robustness evaluated by applying MUSAN noise (5–20 dB SNR) and RIR convolution to evaluation utterances, then re-computing EER.

Table 8. Robustness evaluation under MUSAN noise (5–20 dB SNR).

Condition	EER(%)	minDCF	Notes
Clean (baseline)	49.76	0.9992	No augmentation
Noisy (+MUSAN)	50.06	N/A	Real-world conditions
EER delta	**+0.30**	**—**	**Small degradation**

8. SOFTWARE IMPLEMENTATION

8.1 GitHub Repository

Repository: https://github.com/m24de3067/M24DE3067_SpeakerVerification_Project.git

The repository contains 14 Python source files, 34 unit tests (100% passing), and dataset adapter scripts for VoxCeleb1, CN-Celeb1, and CommonVoice.

8.2 Source Files

Table 9. Source file descriptions and key classes.

File	Purpose	Key Class
src/features.py	Log-Mel Fbank + CMVN	FeatureExtractor
src/model.py	X-Vector & ECAPA models	XVectorModel, ECAPAModel
src/dataset.py	VoxCeleb datasets + fallback	VoxCelebDataset
src/augment.py	MUSAN noise, RIR, speed	AudioAugmenter
src/evaluate.py	EER, minDCF, scoring	Evaluator
src/fairness.py	Per-group EER, bootstrap CI	FairnessAnalyzer
src/explain.py	Integrated Gradients	SaliencyExplainer
src/compare.py	3-model comparison + charts	ModelComparator
src/reweighting.py	WeightedSampler + GRL	FairTrainer
main.py	Pipeline runner (--phase)	main()
tests/test_*.py	34 unit tests — 100% passing	TestFairness, …

8.3 Test Results

Table 10. Unit test results — 34/34 passing.

Test Class	Tests	Coverage
TestFeatureExtractor	5	Fbank/MFCC, CMVN, batch
TestXVectorModel	5	Embedding, gradients, inference
TestECAPAModel	2	Embedding, SE attention
TestAugmentation	3	Noise, RIR, fallback
TestEvaluation	3	EER, minDCF
TestFairness	4	Per-group EER, CI, calibration
TestDatasets	3	Synthetic VoxCeleb, splits
TestDataPrep	3	Trial CSV, score distribution
TestReweighting	4	Sampler, GRL, classifier
TestModelComparison	2	i-vector, comparator
TOTAL	**34**	**100% pass rate**

9. RESPONSIBLE AI FRAMEWORK

Fairness Definition: We adopt Equalized Odds [6] — per-group False Positive Rates are equalised. We acknowledge the fairness impossibility result and report all metrics with 95% bootstrap CIs following NIST evaluation practice.

Transparency: All per-group metrics reported (not only aggregate EER). Codebase and evaluation scripts fully documented and reproducible.

Privacy: Speaker embeddings encode biometric voice identity. Responsible deployment requires embedding de-identification, data minimisation, and GDPR Article 9 compliance.

Limitations:

- Random-weight models: EER ~49% reflects untrained embeddings. Full VoxCeleb2 training expected to yield 3–5% EER with potentially larger absolute gaps.
- Binary categories only: male/female, native/non-native. Age, dialect, emotion not evaluated.
- VoxCeleb1 gender labels from face-recognition metadata — may contain errors.
- Adversarial debiasing implemented but not fully evaluated within project timeline.

10. COMPARISON WITH STATE OF THE ART

Table 11 positions our work against published SOTA on VoxCeleb1. All prior work reports only aggregate EER — fairness analysis columns are empty for every reference paper. Our work is the first to measure, explain, and reduce demographic bias on this benchmark.

Table 11. SOTA comparison on VoxCeleb1. ✓ = only works with fairness analysis. * Random-weight demo.

Method	Venue	EER(%)	Gap	Fairness?
i-vector+PLDA [4]	TASLP'11	~8.00	—	**No**
X-Vectors [2]	ICASSP'18	2.95	—	**No**
VGG-M CNN [8]	IS'17	7.80	—	**No**
ECAPA-TDNN [3]	IS'20	0.87	—	**No**
VoxSRC ensemble [9]	TASLP'24	0.43	—	**No**
Ours — X-Vector	2026	49.06*	0.84%	YES✓
Ours + Reweighting	2026	49.40*	1.23%	YES✓

** EER ~49% reflects random-weight models used for fairness methodology demonstration. Same methodology applies to fully trained models achieving 3–5% EER.*

11. REFERENCES

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